1 Preliminary Knowledge

1.1 Quadrature Operator

We define the quadrature operator $\hat{X}_\theta$ as

$$\hat{X}_\theta = \hat{a}e^{-i\theta} + \hat{a}^\dagger e^{i\theta} \tag{1}$$

The conventional field quadratures used in quantum optics are $\hat{X} = \hat{X}_0$ and $\hat{P} = \hat{X}_{\pi/2}$

$$\hat{X} = \hat{a} + \hat{a}^\dagger, \quad \hat{P} = (\hat{a} - \hat{a}^\dagger)/i \tag{2}$$

and satisfy the commutation relation

$$[\hat{X}, \hat{P}] = 2i, \tag{3}$$

building upon the fact that $[\hat{a}, \hat{a}^\dagger] = 1$. These field quadratures define the Wigner function plane.

1.2 Fock States

In the Fock basis representation, Fock (or number) states are denoted as $|n\rangle$, where n is the number of photons (or excitations) in the vacuum $|0\rangle$. They are eigenstates of the number operator

$$\hat{n}\,|n\rangle = n\,|n\rangle\,, \quad (\hat{n} = \hat{a}^\dagger \hat{a}). \tag{4}$$

As eigenstates of the number operator, they have a well-defined number of photons $(\Delta\hat{n})^2 = \langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2 = 0$, but an undefined optical phase ϕ. Their wigner function is a gaussian centered at the origin.

In the position basis representation, a fock state is written as

$$\psi_n(x) = \langle x|n\rangle = \frac{1}{\pi^{1/4}\sqrt{2^n n!}} H_n(x) e^{-x^2/2}, \tag{5}$$

where $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$ is the Hermite polynomial of order n.

1.3 Coherent States

Coherent states are defined as the eigenstates of the annihilation operator

$$\hat{a}\,|a\rangle = a\,|a\rangle\,, \quad a \in \mathbb{C}. \tag{6}$$

The complex amplitude of the state can be written as $\langle \hat{a} \rangle = a = |a|e^{i\phi}$, where ϕ is the optical phase of the electromagnetic field. In the Fock basis, a coherent state is decomposed as

$$|a\rangle = e^{-\frac{|a|^2}{2}} \sum_{n=0}^{\infty} \frac{a^n}{\sqrt{n!}} |n\rangle\,. \tag{7}$$

The average number of photons in a coherent state is

$$\langle a|\hat{n}|a\rangle = |a|^2. \tag{8}$$

Such state can be generated through displacement of the vacuum

$$|a\rangle = \hat{D}(a)\,|0\rangle\,, \quad \hat{D}(a) = \exp\left[a\hat{a}^\dagger - a^*\hat{a}\right]. \tag{9}$$

In our convention, the Wigner function of a coherent state is displaced from the origin by an amount $2|a|$ and at an angle ϕ from the x-axis.

1.4 Squeezed States

Squeezed states are generated through squeezing of the vacuum

$$|z\rangle = \hat{S}(z)\,|0\rangle\,, \quad \hat{S}(z) = \hat{S}(z) = \exp\left[\frac{1}{2}(z*\hat{a}^2 - z\hat{a}^{\dagger 2})\right]. \tag{10}$$

The complex number $z = |r|e^{i\theta}$ determines the squeezing parameter r and the squeezed quadrature $\hat{X}_\theta$. Squeezing the quadrature $\hat{X}_\theta$ results in the anti-squeezing of quadrature $\hat{X}_{\theta+\pi/2}$

$$(\Delta\hat{X}_\theta)^2 \propto e^{-2r}, \quad (\Delta\hat{X}_{\theta+\pi/2})^2 \propto e^{2r}. \tag{11}$$

The average photon number of a squeezed state is

$$\langle z|\hat{n}|z\rangle = \sinh^2(r). \tag{12}$$

The Wigner function is elliptical due to the squeezing, and at an angle $\theta/2$ with respect to the x-axis.

1.5 Displaced Squeezed States (DSS)

Such states are generated through squeezing and displacement of the vacuum

$$|a,z\rangle = \hat{D}(a)\hat{S}(z)\,|0\rangle\,. \tag{13}$$

Previous points for displaced and squeezed states hold independently. The average number of photons in a DSS is

$$\langle a,z|\hat{n}|a,z\rangle = |a|^2 + \sinh^2(r). \tag{14}$$

The Wigner function is again elliptical, displaced from the origin by an amount $2|a|$, is at an angle ϕ with respect to the x-axis, and rotated by $\theta/2$.

2 Discrimination between pure Quantum States in a communication channel

The minimum theoretical error probability of mis-discriminating between two non-orthogonal pure states $|\psi_{1,2}\rangle$ (with equal prior probability of being selected) is given by the Helstrom bound

$$P^{(\mathrm{min})}(|\psi_1\rangle, |\psi_2\rangle) = \frac{1}{2}\left(1 - \sqrt{1 - |\langle\psi_1|\psi_2\rangle|^2}\right) \tag{15}$$

Experimentally, one cannot reach the absolute minimum probability of making an error when trying to discriminate between two states. In this work, Homodyne detection is considered as the experimental means of a receiver to discriminate between two states, which clearly leads to a higher error probability. The strategy used by the receiver is the following:

$$\begin{cases} x \geq 0 \Rightarrow |0\rangle \equiv |\psi_1\rangle\,, \\ x < 0 \Rightarrow |1\rangle \equiv |\psi_2\rangle \end{cases} \tag{16}$$

where x is drawn from the Homodyne detection probability distribution

$$x \sim P^{(\mathrm{HD})}(x; |\psi_{1,2}\rangle) = |\langle x|\psi_{1,2}\rangle|^2, \tag{17}$$

with $\hat{X}\left|x\right\rangle = x\left|x\right\rangle$. The probability of making an error is given by

$$P_{\text{err}} = \frac{1}{2}[p(x \geq 0|x=1) + p(x<0|x=0)] = p(x \geq 0|x=1); \quad p(x \geq 0|x=1) = p(x<0|x=0) \tag{18}$$

The conditional probability is given by

$$p(x \geq 0|x=1) = \int_0^\infty P^{(\text{HD})}(x; |\psi_{1,2}\rangle)dx. \tag{19}$$

To approximate this probability, we draw thousands of samples from $P^{(\text{HD})}(x; |\psi_{1,2}\rangle)$, and define the error probability as

$$p(x \geq 0|x=1) = P_{\text{err}} = \frac{\text{\# of samples where x}\geq 0}{\text{\# of all samples}} \tag{20}$$

$$p(x < 0|x=0) = P_{\text{err}} = \frac{\text{\# of samples where x}<0}{\text{\# of all samples}} \tag{21}$$

The goal of the work is to determine whether the application of certain combinations of quantum resources on the states (e.g. displacement, squeezing) can make $P^{(\text{HD})}(x; |\psi_{1,2}\rangle)$ approach $P^{(\text{min})}(|\psi_1\rangle, |\psi_2\rangle)$ as much as possible with respect to other combinations.

2.1 Discrimination between Coherent States

In the case of coherent state communication our states are defined as

$$|\psi_k\rangle = \hat{D}\left[(-1)^k a\right]|0\rangle, \quad k = 1,2 \tag{22}$$

where $a \in \mathbb{R}$ and $k = 1,2$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P_{CS}^{(\text{min})}(N) = \frac{1}{2} - \frac{1}{2}\sqrt{1 - \exp(-4N)}, \tag{23}$$

where N is the average number of photons in the coherent states. The homodyne detection probability distribution becomes

$$P_{CS}^{(\text{HD})}(x; |\pm a\rangle) = \frac{1}{2}\text{erfc}\left(\sqrt{2N}\right). \tag{24}$$

2.2 Discrimination between DSS

In the case of communication with DSS, our states are defined as

$$|\psi_k\rangle = \hat{D}\left[(-1)^k a\right]\hat{S}(r)|0\rangle, \quad k = 1,2 \tag{25}$$

where $z = r \in \mathbb{R}$ and $k = 1,2$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P^{(\text{min})}(N) = \frac{1}{2} - \frac{1}{2}\sqrt{1 - \exp(-4Ne^{-2r})}, \tag{26}$$

where N is the average number of photons in the DSS

$$N = |a|^2 + \sinh^2(r). \tag{27}$$

We may define the squeezing fraction $\beta \equiv \sinh^2(r)/N$, and express the relevant equations with respect to that. The minimum error probability (Helstrom Bound) is given by

$$P_{DSS}^{(\min)}(N,\beta) = \frac{1}{2} - \frac{1}{2}\sqrt{1 - \exp\left[-4N(1-\beta) \times \left(1 + 2N\beta + 2\sqrt{N\beta(1+N\beta)}\right)\right]}, \tag{28}$$

The homodyne detection probability distribution becomes

$$P_{DSS}^{(\mathrm{HD})}(x;|\pm a\rangle) = \frac{1}{2}\mathrm{erfc}\left(\sqrt{2}\frac{\alpha}{\Sigma}\right), \tag{29}$$

where

$$\alpha = \sqrt{N(1-\beta)}, \quad \Sigma = \left(\sqrt{N\beta} + \sqrt{1+N\beta}\right) \tag{30}$$

2.3 Which is easier to discriminate, Coherent states or DSS?

For a certain fraction of squeezing, DSS can outperform Coherent states. More specifically, for a fixed number of photons $\tilde{N}$, DSS have a lower error probability if β is below a threshold β_{th}

$$P_{\mathrm{err}}^{DSS}(\beta|\tilde{N}) \leq P_{\mathrm{err}}^{CS}(\tilde{N}) \quad \text{if } \beta \leq \beta_{\mathrm{th}}(\tilde{N}), \tag{31}$$

where

$$\beta_{\mathrm{th}}(\tilde{N}) = \frac{4\tilde{N}}{4\tilde{N}+1}. \tag{32}$$

Additionally, for a fixed number of photons $\tilde{N}$, there exists an optimal fraction of squeezing $\beta_{\mathrm{opt}} \in [0, \beta_{\mathrm{th}}]$ for which the error probability is minimized

$$\left.\frac{\partial P_{\mathrm{err}}^{DSS}}{\partial \beta}\right|_{\beta_{\mathrm{opt}},\tilde{N}} = 0 \Rightarrow \beta_{\mathrm{opt}} = \frac{\tilde{N}}{2\tilde{N}+1}. \tag{33}$$

Summarizing, for a fixed number of photons $\tilde{N}$, DSS will always outperform Coherent states if the fraction of squeezing is lower than $\beta_{\mathrm{th}}(\tilde{N})$. Moreover, the error probability will be minimized if the fraction of squeezing is equal to $\beta_{\mathrm{opt}}(\tilde{N}) < \beta_{\mathrm{th}}(\tilde{N})$.

3 Discrimination between Quantum States under phase diffusion in a communication channel

We can describe the effect of phase diffusion on the state through the following differential equation for the density matrix

$$\frac{1}{\Gamma}\frac{d\rho(t)}{dt} = 2\hat{n}\rho\hat{n}^\dagger - \hat{n}^\dagger\hat{n}\rho - \rho\hat{n}^\dagger\hat{n}; \tag{34}$$

we may write the density operator in the Fock basis by using the resolution of identity

$$\rho = \mathbb{I}\rho\mathbb{I} = \left(\sum_n |n\rangle\langle n|\right)\rho\left(\sum_m |m\rangle\langle m|\right), \tag{35}$$

$$= \sum_{n,m} \langle n|\rho|m\rangle |n\rangle\langle m|, \tag{36}$$

$$= \sum_{n,m} \rho_{n,m} |n\rangle\langle m|. \tag{37}$$

Then, eq. (34) becomes

$$\frac{1}{\Gamma}\frac{d\rho(t)}{dt} = 2\sum_{n,m} nm\rho_{n,m}\left|n\right\rangle\left\langle m\right| - \sum_{n,m} n^2\rho_{n,m}\left|n\right\rangle\left\langle m\right| - \sum_{n,m} m^2\rho_{n,m}\left|n\right\rangle\left\langle m\right|, \tag{38}$$

$$\Rightarrow \sum_{n,m}\left[\frac{1}{\Gamma}\dot{\rho}_{n,m} - 2nm\rho_{n,m} + (n^2+m^2)\rho_{n,m}\right]\left|n\right\rangle\left\langle m\right| = 0, \tag{39}$$

$$\Rightarrow \dot{\rho}_{n,m} + \Gamma(n-m)^2\rho_{n,m} = 0. \tag{40}$$

Finally,

$$\rho_{n,m}(t) = e^{-(n-m)^2\Gamma t}\rho_{n,m}(0); \tag{41}$$

the full density matrix is

$$\rho(t) = \sum_{n,m}\rho_{n,m}(t)\left|n\right\rangle\left\langle m\right| = \sum_{n,m} e^{-(n-m)^2\Gamma t}\rho_{n,m}(0)\left|n\right\rangle\left\langle m\right|. \tag{42}$$

By the use of Fourier transform theory, we can rewrite the above equation as

$$\rho(t) = \int_{-\infty}^{+\infty}\left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}\right)\hat{U}_\phi\rho(0)\hat{U}_\phi^\dagger d\phi, \tag{43}$$

where $\hat{U}_\phi = e^{-i\hat{n}\phi}$ and $\sigma \propto \sqrt{\Gamma}$ is the noise parameter. This operator has the following effect on Coherent states and DSS

$$\hat{U}_\phi\left|\pm a\right\rangle = e^{-i\hat{n}\phi}\left|\pm a\right\rangle = \left|\pm e^{-i\phi}a\right\rangle, \tag{44}$$

$$\hat{U}_\phi\left|\pm a, r\right\rangle = e^{-i\hat{n}\phi}\left|\pm a, r\right\rangle = \left|\pm e^{-i\phi}a, e^{-i2\phi}r\right\rangle. \tag{45}$$

Evidently, this operator produces an optical phase rotation. So, for both Coherent and DSS, the density matrix of eq. (43) is

$$\rho^{CS,DSS}(t) = \int_{-\infty}^{+\infty}\left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}\right)\times\left(\rho^{CS,DSS}_{\text{phase-shift}}\right)d\phi \tag{46}$$

$$= \int_{-\infty}^{+\infty} P(\phi)\times\left(\rho^{CS,DSS}_{\text{phase-shift}}\right)d\phi \tag{47}$$

Thus, **the state is a statistical mixture of phase-rotated Coherent states or DSS, where the phase is drawn from a Gaussian distribution with variance σ^2.** This is not the same as having a superposition of phase-rotated states; the state has undergone only one specific phase rotation - no one knows which exactly, this is what makes the state mixed. Thus, to reproduce the state in simulation, one needs to create a pure Coherent state / DSS, and then apply a phase rotation, where the phase is drawn from a Gaussian distribution with mean zero and variance σ^2

$$\rightarrow \text{create desired state } \left|\psi\right\rangle, \tag{48}$$

$$\rightarrow \text{pick } \phi \sim \mathcal{N}(0,\sigma^2), \tag{49}$$

$$\rightarrow \text{apply phase shift operator } \hat{U}_\phi\left|\psi\right\rangle. \tag{50}$$

Then, Homodyne measurement is carried out as originally.

3.1 Discrimination between phase diffused Coherent states

In the case of coherent state communication our states are defined as

$$\rho_k = \int_{-\infty}^{+\infty} \left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \right) |(-1)^k e^{-i\phi} a\rangle \langle (-1)^k e^{-i\phi} a| \, d\phi, \quad k = 1,2 \,, \tag{51}$$

where $a \in \mathbb{R}$ and $k = 1,2$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P_{CS}^{(\min)}(\rho_1, \rho_2) = \frac{1}{2} \left[1 - \frac{1}{2} \mathrm{Tr} \left[\sqrt{(\rho_1 - \rho_2)^\dagger (\rho_1 - \rho_2)} \right] \right]. \tag{52}$$

The Homodyne detection probability distribution for x is

$$P_{CS}^{(\mathrm{HD})}(x| |\pm e^{i\phi} a\rangle) = \int_{-\infty}^{+\infty} P(\phi) |\langle x| \pm e^{i\phi} a\rangle|^2 d\phi, \tag{53}$$

$$= \int_{-\infty}^{+\infty} \frac{P(\phi)}{\sqrt{2\pi}} \exp\left[-\frac{(x \mp 2a\cos\phi)^2}{2} \right] d\phi, \tag{54}$$

$$= \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(x \mp 2a\cos\phi)^2}{2} \right] d\phi. \tag{55}$$

3.2 Discrimination between phase diffused DSS

In the case of communication with DSS our states are defined as

$$\rho_k = \int_{-\infty}^{+\infty} \left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \right) |(-1)^k e^{-i\phi} a, e^{-i2\phi} r\rangle \langle (-1)^k e^{-i\phi} a, e^{-i2\phi} r| \, d\phi, \quad k = 1,2 \,, \tag{56}$$

where $z = r \in \mathbb{R}$ and $k = 1,2$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P_{DSS}^{(\min)}(\rho_1, \rho_2) = \frac{1}{2} \left[1 - \frac{1}{2} \mathrm{Tr} \left[\sqrt{(\rho_1 - \rho_2)^\dagger (\rho_1 - \rho_2)} \right] \right]. \tag{57}$$

The Homodyne detection probability distribution for x is

$$P_{DSS}^{(\mathrm{HD})}(x| |\pm e^{i\phi} a, e^{-i2\phi} r\rangle) = \int_{-\infty}^{+\infty} P(\phi) |\langle x| \pm e^{i\phi} a, e^{-i2\phi} r\rangle|^2 d\phi, \tag{58}$$

$$= \int_{-\infty}^{+\infty} \frac{P(\phi)}{\sqrt{2\pi V(\phi)}} \exp\left[-\frac{(x \mp 2a\cos\phi)^2}{2V(\phi)} \right] d\phi, \tag{59}$$

$$= \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \frac{1}{\sqrt{2\pi V(\phi)}} \exp\left[-\frac{(x \mp 2a\cos\phi)^2}{2V(\phi)} \right] d\phi, \tag{60}$$

where

$$V(\phi) = e^{-2r} \cos^2\phi + e^{2r} \sin^2\phi. \tag{61}$$